A Cross-layer triangulation on the frozen NCP topology

Layer-specific summaries are juxtaposed without fitting a merged latent model.

FROZEN BACKBONE
16 frozen NCP families
Canonical membership, direction & display order from Fig 1D
HCA LA carrier prior: context only

PAIRED-CHAMBER BULK
Cross-cohort consistency
Dominant rewiring state
Path A gate: High consistency
Dominant state: context only

LA SINGLE-NUCLEUS
Fidelity
Relay
Extension
Below-detection
Path A gate: non-Below-detection

LAA DURATION
Strong-trend
Weak-trend
Stable
Path A gate: non-Stable
Class uses unrounded $|\rho_{LAA}|$

RULE-BASED TRIAGE
Three Path A gates
Bulk High | snRNA non-Below | Duration non-Stable
support count = 0–3
Path A = 3/3 gates
Rule-based prioritization
Not a unified significance test or causal validation

GeoMx spatial context · 24 eligible ROIs · 4 patients · ROI-level only · not included in support count or Path A · shown in D

B Family-level cross-layer triangulation matrix

Context annotations and three Path A gates in the frozen Fig 1D order
HCA carrier and bulk dominant state are context only; support count uses bulk, snRNA and duration gates.

			CONTEXT		PATH A GATES			TRIAGE	
			HCA LA carrier	Bulk dominant state	Bulk consistency (gate: High)	LA snRNA class (gate: non-Below-detection)	LAA duration class (gate: non-Stable)	Support count	Path A
LA	F1	Translation / RQC	Endo	Rev	H	Below	Stable	1/3	–
LA	F2	Mito Ca^{2+} transport	aCM	Att	M	Below	Strong	1/3	–
LA	F3	Adipo–lipid axis	Adipo	Pres	M	Below	Weak	1/3	–
LA	F4	Mito proteostasis / FA–BCAA	Adipo	Att	L	Below	Stable	0/3	–
LA	F5	tRNA aminoacylation	Lymph	Pres	M	Below	Stable	0/3	–
LA	F6	Mito translation / import	Mixed	Rev	M	Below	Stable	0/3	–
LA	F7	OXPHOS / ETC	Mixed	Rev	L	Below	Weak	1/3	–
LA	F8	Complex I biogenesis	Endo	Rev	M	Below	Weak	1/3	–
LA	F12	rRNA processing	Mixed	Pres	H	Below	Stable	1/3	–
LA	F13	Amino-acid metabolism	Endo	Pres	M	Below	Weak	1/3	–
LA	F14	Translation initiation	Endo	Rev	H	Below	Stable	1/3	–
RA	F9	HS-GAG / bone / GPCR	Fibro	Att	H	Below	Strong	**2/3**	–
RA	F10	GAG disease / TGF-β	Fibro	Att	H	Fidelity	Weak	**3/3**	**Path A**
RA	F11	BMP signalling	Endo	Pres	M	Relay	Stable	1/3	–
RA	F15	NTRK2 / TrkB	Fibro	Att	H	Relay	Weak	**3/3**	**Path A**
RA	F16	Keratan sulfate	Fibro	Att	M	Extension	Stable	1/3	–

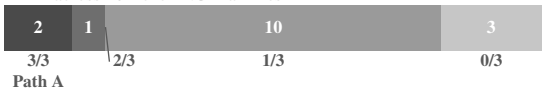
Legend

Bulk consistency: High · Moderate · Low
Bulk state: Amplified · Preserved · Attenuated · Reversed
LA snRNA: Fidelity · Relay · Extension · Below-detection
LAA duration: Strong · Weak · Stable

Context columns do not enter support count. Path A requires all three gate columns.

C Rule-based Path A prioritization

Observed gate combinations
across 16 frozen NCP families



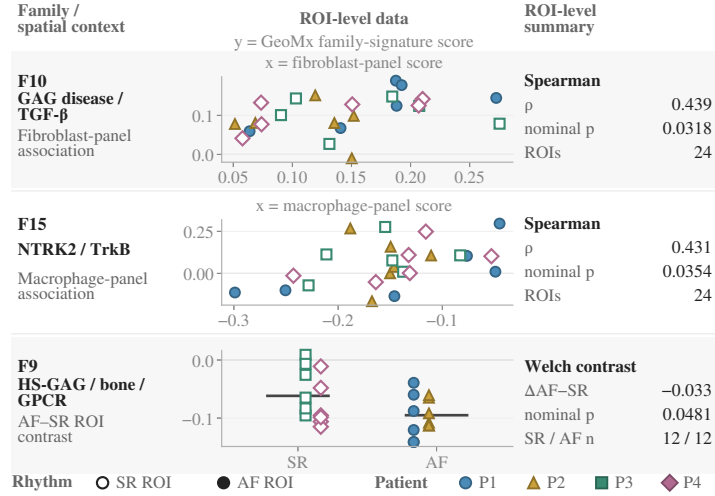
Bulk High	LA snRNA gate: non-Below-detection	LAA duration non-Stable	n	Families
●	●	●	2	F10 · F15
●	○	●	1	F9
●	○	○	3	F1 · F12 F14
○	●	○	2	F11 · F16
○	○	●	5	F2 · F3 F7 · F8 F13
○	○	○	3	F4 · F5 F6

Path A F10 GAG disease / TGF-β F15 NTRK2 / TrkB

Context annotations, GeoMx, and genetics are not counted.
● gate met; ○ gate unmet. Path A requires 3/3 gates.

D GeoMx ROI spatial context

24 eligible cardiac AA DNA-positive ROIs
· 4 patients · descriptive context only



Patient = color/shape; rhythm = open/filled; panel-specific axes.
ROI-level summaries; within-patient ROI clustering was not modeled.